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6 Summary and future trends

The previous chapters in this book confirm that, since the 1950s when the age of jet-powered civil aviation began, the superalloys have underpinned the improvements in performance of the modern gas turbine engine [1]. Today's designs are appreciably superior to the Ghost engine used for the first commercial flight of the Comet I; see Figure 6.1. The modern civil turbofans which power large two-engined aircraft such as the Boeing 777 at a take-off thrust approaching 100 000 lb – for example, Rolls-Royce's Trent 800 or General Electric's GE90 – have a fuel economy which has improved by a factor of about 2; the engine weight, when normalised against the thrust developed, is lower by about a factor of 4. Whilst a number of factors have contributed to this technological success, the development of the superalloys and insertion of components fabricated from them into the very hottest parts of the turbine have been absolutely critical to it. As the compositions of the alloys have evolved, the critical properties in creep and fatigue have improved markedly – and this has allowed the turbine entry temperature to be increased beyond 1700 K; see Figure 1.5. The development of new processes has played a key role too, particularly vacuum melting, directional solidification/investment casting and powder metallurgy techniques.

It is instructive to consider the technological, economic and societal pressures which have provided the incentive for these developments. In the post-war period to the late 1970s when the Cold War was at its height, most cutting-edge superalloy development was carried out in support of engine designs for military aircraft – notably in the USA, Europe and the former USSR. During this time, new superalloy and coatings technology was adopted first in fighter aircraft – with military organisations paying for it via government subsidies. With the demise of the Iron Curtain, the situation altered in a subtle way: the civil rather than the military sector set the pace, with new materials being inserted first into new civil aeroengines. In the 1990s, the need for cost reduction was apparent as the civil aviation sector became more competitive; the aeroengine business responded with emphasis being placed on 'faster and cheaper' derivative designs, on account of the variety of airframes becoming available and the speed at which these were being designed and marketed. At this time, increased use was made of computer-based design for alloy development and component optimisation via processing modelling. Around the turn of the century, substantial growth in the industrial gas turbine (IGT) business was observed – due to the 'dash-for-gas', notably in Europe and an increased thirst for electricity in the developing world, particularly in China and India. For example, in 1999 sales of IGTs increased from \$8.5 bn to \$13.0 bn, and the total share of the gas turbine market increased from 30% to 38% compared with the previous year;

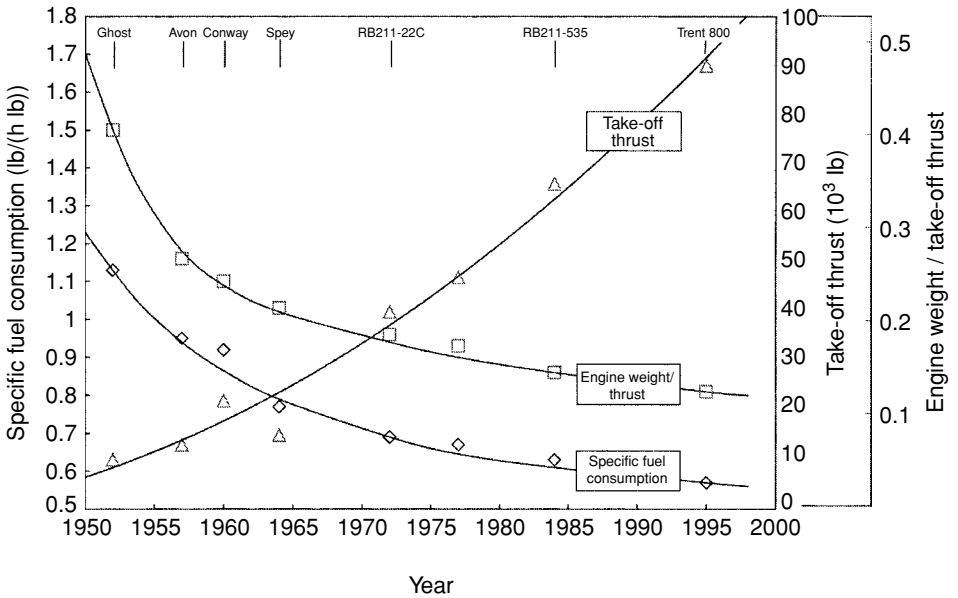


Fig. 6.1. Trends in the performance characteristics (specific fuel consumption, take-off thrust and normalised engine weight) for Rolls-Royce's large civil aeroengines.

by contrast, the aeroengine market share declined from 65% to 59%, despite an absolute increase in sales from \$18.0 bn to \$20.0 bn [2]. Although the downturn in the first few years of the new millennium and other unforeseen circumstances meant that some of this optimism was misplaced in the short term, most projections indicate that by 2025 the market for IGTs will outstrip that for aeroengines, which is itself projected to double by that time. In part this is due to an increased reliance on combined-cycle power plant, which requires IGTs to be used in conjunction with steam-powered turbines. For intermediate-sized plant in the range 25 to 75 MW, there is a growing market for aeroderivative IGTs – particularly for the oil and gas business in places such as Azerbaijan, Brazil, China, Kazakhstan and West Africa – which make use of materials technologies designed originally for aeroengine applications; it is anticipated that these will constitute about 10% of the total IGT market by 2025. For both aeroengines and IGTs, an increasingly important driver will be environmental regulations, introduced as the world's supply of oil and gas becomes tighter and concerns grow over the effects of CO₂ emissions [3].

So what of the future? Whatever the balance of power between the aeroengine and industrial gas turbine sectors, it seems certain that further developments in superalloy metallurgy and processing will be required over the coming decades. In the aeroengine sector, for example, it is projected that over the next 20 years a market exists for about 90 000 new civil engines, worth an estimated \$550 bn; much of this requirement is driven by the rapid growth anticipated in Asia, the replacement of about 10 000 ageing aircraft, a booming business jet market and the trend for low-cost airlines. The required improvements in performance, for example, for the Trent 1000 and GENx for the Boeing 787 Dreamliner, are ambitious; new grades of superalloy and supporting technologies are required to bring these new designs

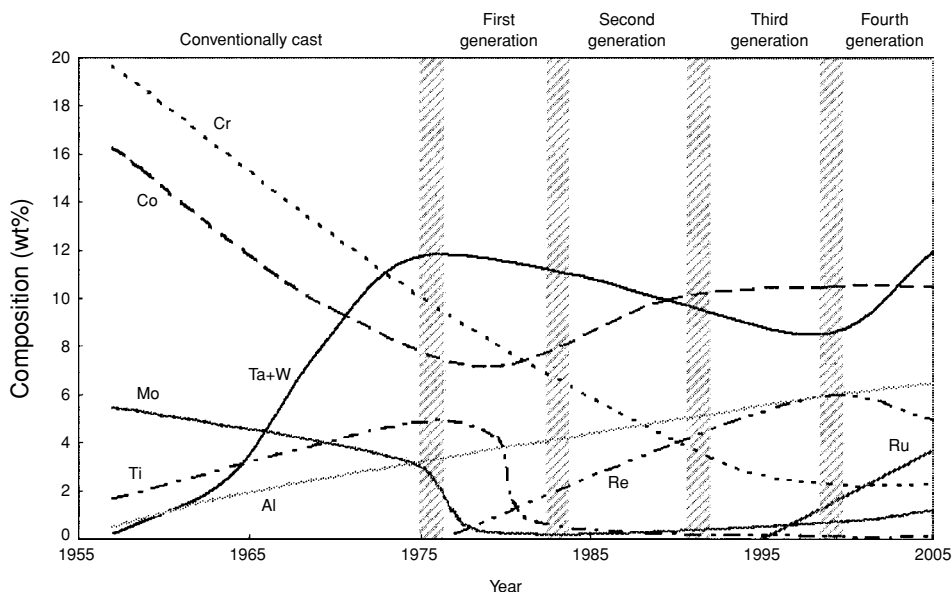


Fig. 6.2. Evolution of the compositions of the single-crystal superalloys for high-pressure turbine blading, to 2005.

to fruition. Similarly, as the operating conditions of land-based turbines become more demanding, superalloys will be used increasingly for hot section components such as turbine blading and buckets.

6.1 Trends in superalloys for turbine blade applications

Consider the evolution of single-crystal turbine blade compositions for the high-pressure turbine blading of the aeroengines. Figure 6.2 provides an approximate overview – note some subjectivity is inherent in it since, at any given time, different organisations will have been working on different alloys to exploit patent positions. The reasoning behind the changes can be rationalised given the findings of Chapter 3. Since creep resistance has been imparted by raising the γ' content of the alloys to 75%, the concentration of the γ -forming element, Cr, has dropped markedly; the Al content has been increased to achieve this, it being a strong γ' former. These modifications have, in turn, promoted changes in the oxidation behaviour – in the uncoated condition, most modern alloys favour Al_2O_3 rather than Cr_2O_3 scales; thus no great resistance to hot corrosion is displayed by them and appropriate coating technologies need to be adopted where necessary, for example, in marine environments. In many alloys, titanium is now present at very low levels since it exacerbates the rate of oxidation and has a negative effect on the retention of thermal barrier coatings (TBCs). No strong consensus has been reached on the role of Co, and consequently wide variations in its concentration are found. In the early first-generation single-crystal alloys, Ta and W additions were included to promote creep strengthening, but their use later

declined as other more potent strengthening elements, for example, Re, were identified. In the second-generation single-crystal superalloys, Re was added at levels of about 3 wt%, in view of its potent creep strengthening effect. Concentrations of Re were subsequently increased to 6 wt% in the third-generation alloys to exploit further the so-called 'Re-effect'.

It is now recognised that the third-generation superalloys – which are heavily alloyed with Re – are unstable with respect to the formation of topologically close-packed (TCP) phases; these form under aluminide coatings within secondary reaction zones (SRZs), within Re-rich dendritic regions (where casting microsegregation persists) and at low angle boundaries. The rupture life is impaired when TCP precipitation occurs; see Figure 6.3. This is an issue for high-pressure turbine blading since the cross-sectional area of the thin-walled sections, which is effective at load-bearing, is reduced by such instability. Recent research has demonstrated that Ru reduces the susceptibility to TCP formation (see Section 3.2), although the reasons for this are not yet fully understood; consequently, the fourth-generation superalloys developed since the turn of the century are characterised by additions of Ru but with Re levels lower than for the third-generation alloys. Concentrations of Re at 4.5 wt% and Ru at 4.0 wt% are typical for these new fourth-generation alloys; see Table 3.5. An alternative strategy is to make use of platinum-diffused rather than aluminide-type diffusion coatings; these are not prone to the SRZ effect, although TCP formation is still possible in the bulk. At the time of writing, these fourth-generation single-crystal superalloys have yet to find application in gas turbine engines, although their potential benefit is acknowledged. The current reluctance to insert them into the latest designs is due to their cost and density; see Figure 6.4. A strong correlation exists between creep rupture life and cost, largely on account of the significant fractions of Re and Ru in the latest alloys; it can be seen that the very best creep resistance is displayed by alloys with raw material costs of about five times that of the first-generation alloys. A further difficulty is that the latest generation superalloys have densities approaching, or even exceeding, 9 g/cm^3 – partly because they are still alloyed heavily with Re, albeit at lower levels than for the third-generation superalloys. The need to limit the density is acute since the loading at the blade root scales linearly with it and because stresses imparted on the disc rim are then enhanced. Any fourth- or fifth-generation single-crystal superalloy that is to find application in the gas turbine engine will need to balance properly the very evident trade-off between creep performance, density and cost.

Traditionally, superalloys for turbine blade applications have been designed with creep, fatigue and oxidation resistance in mind. Whilst these are still very important, a new requirement of coating compatibility is now emerging, since the turbine entry temperature (TET) of the modern aeroengine is outpacing the temperature capability of the single-crystal superalloys, for which the melting temperature of nickel represents an absolute ceiling; see Figure 1.5. Hence, the use of thermal barrier coatings for protecting high-pressure blading is no longer merely advisable but absolutely critical, and consequently an increasing number of rows of aerofoils are employing cooling air; use of TBCs is also spreading to non-rotating parts such as nozzle guide vanes. This explains the great emphasis being placed on newer, lower-conductivity TBCs, ones which display resistance to sintering, and on the commercial aspects of TBC development including reduced manufacturing costs. Hence a new role for the superalloy blading relates to its requirement to support the TBC throughout service without spallation occurring; this, in turn, is focussing attention on the interfacial effects

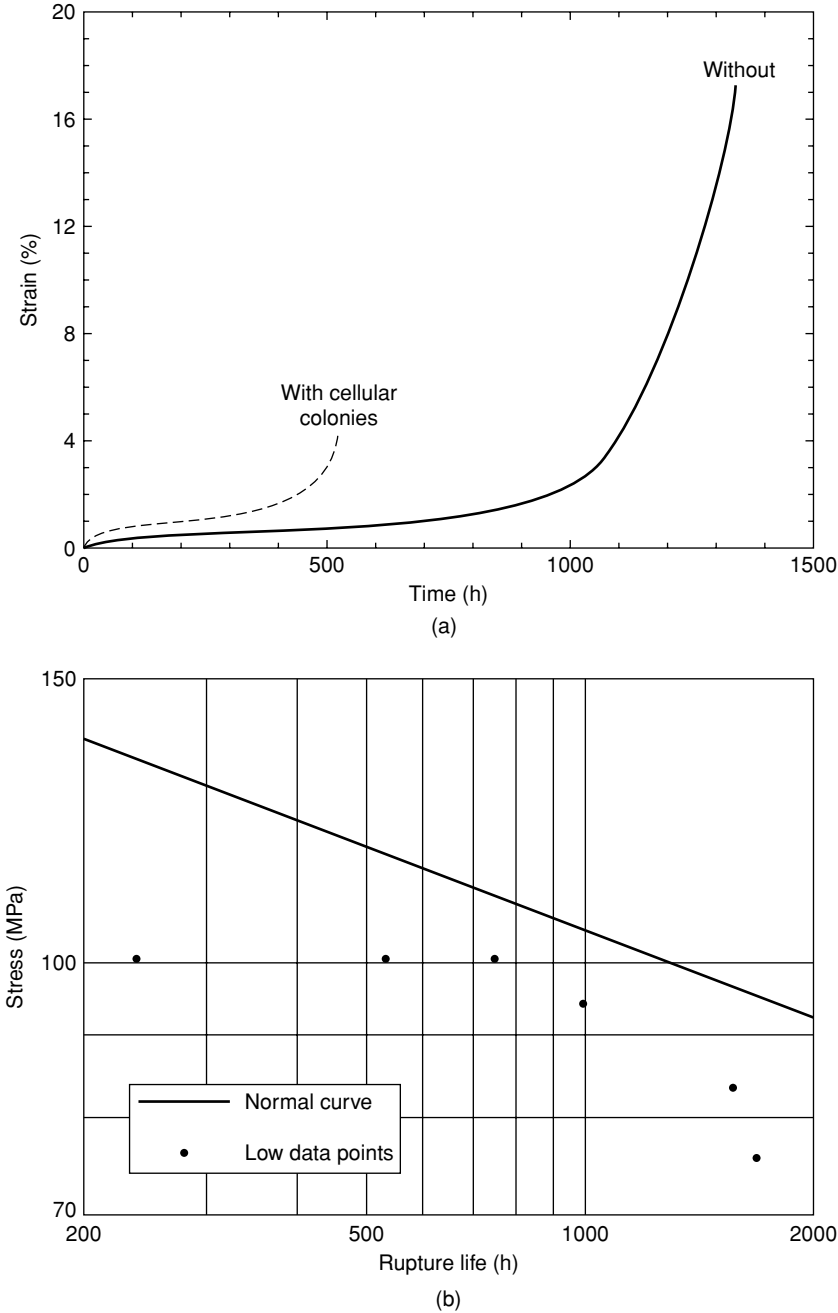


Fig. 6.3. Data confirming the debit in creep rupture life when third- and fourth-generation superalloys are tested with cellular colonies containing TCP phases. (a) Creep curves showing the detrimental effect of the presence of cellular colonies (1093 °C, 103 MPa); (b) creep rupture data confirming the reduced lifetime as a function of stress applied (1093 °C). (Courtesy of Scott Walston.)

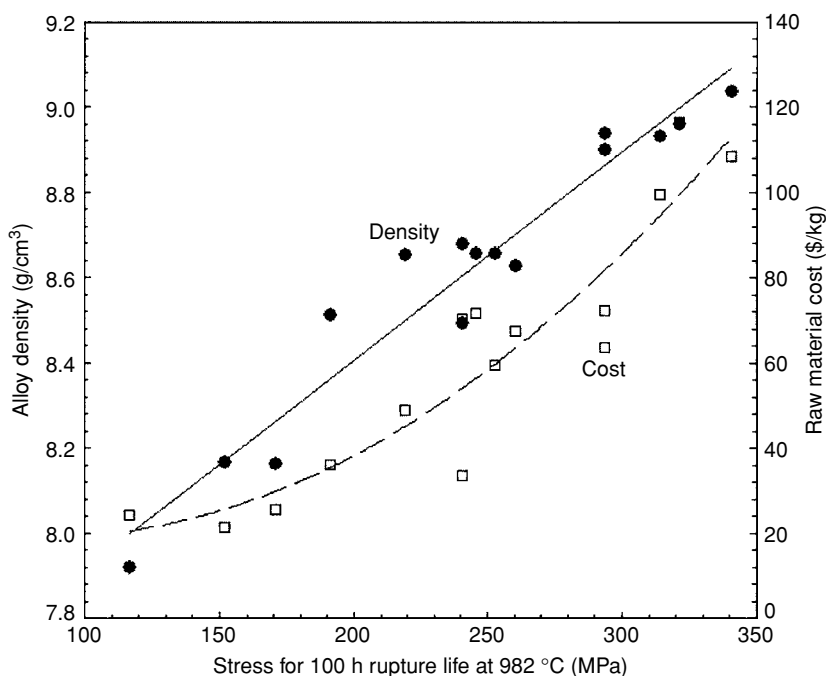


Fig. 6.4. Data for the density, cost and associated creep performance of a number of third- and fourth-generation single-crystal superalloys, illustrating that improved creep capability is won only by increasing the density (by including heavy elements such as Re) and therefore also the cost. (Courtesy of Scott Walston.)

which control whether TBC adhesion can be maintained for the whole of the service life. It is known that different superalloy substrates display different TBC spallation lives, even for identical bond coat/top coat combinations and processing conditions; this emphasises that chemical effects are inherited from the substrate. Interestingly, it appears that there exists no strong correlation between the oxidation resistance of an uncoated alloy and the ability of a TBC system to adhere to it over many heating/cooling cycles. These observations emphasise that the turbine blading, the bond coat and the ceramic top coat can no longer be designed in isolation. A so-called ‘systems-based’ approach will be necessary. This will be a fruitful area for future research.

Given these considerations, it will be appreciated that it is becoming progressively more difficult to design new ‘next-generation’ single-crystal superalloy compositions to satisfy simultaneously the many requirements of a turbine blade: low density, stability with respect to TCP formation, oxidation resistance, corrosion resistance, coating compatibility, castability, low cost and low/high temperature creep resistance. This is partly due to the ‘extra’ requirements of cost, density and coating compatibility, which have become more apparent in recent years, but which traditionally were not emphasised strongly. Also of relevance is that any one alloying element is unlikely to improve all properties simultaneously, for example, Ru improves stability but increases cost, Re improves high-temperature strength but increases density, Cr improves corrosion resistance but degrades stability, etc.

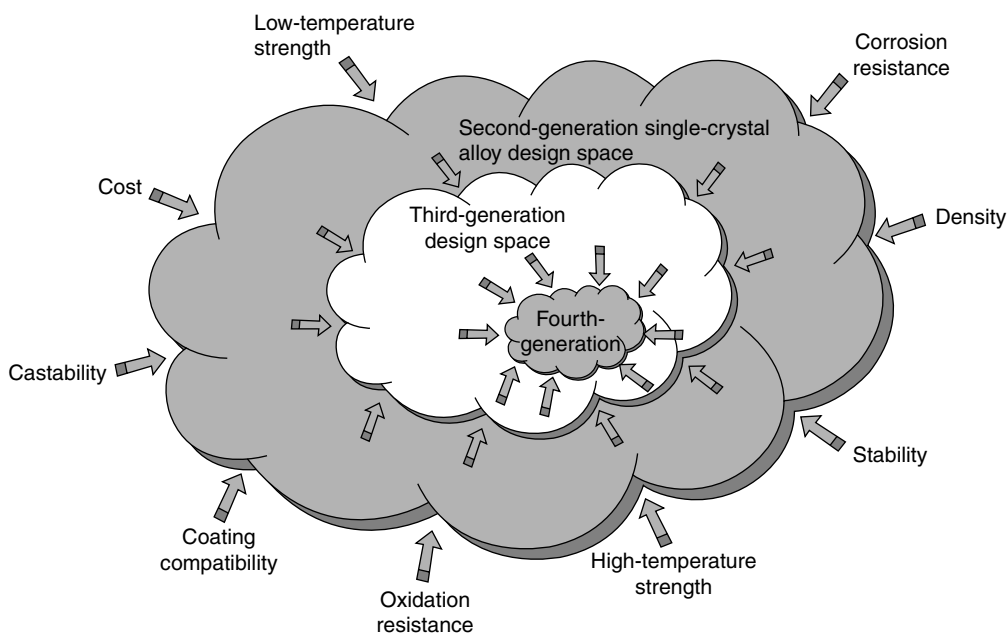


Fig. 6.5. Schematic illustration of the restricted size of the 'design space' available after the composition of each new generation of single-crystal superalloy has been chosen. (Courtesy of Scott Walston.)

Walston has illustrated this difficulty in a pictorial sense (see Figure 6.5), pointing out that the 'design space' available for next-generation single-crystal superalloys is becoming increasingly 'crowded'. From the turbine designers' perspective, this means that decisions need to be made concerning the relative importance of the requirements to be exhibited. It means also that it is unlikely that a 'one-size-fits-all' composition will be identified which is suitable for all turbine engines, so that there is scope for compositions 'tailored' for different applications. For example, the metallurgist designing a superalloy composition for an IGT is likely to place greater emphasis on corrosion resistance, cost and castability due to the large size of the blading and the operating environment [3]; retention of material properties, and thus microstructural stability, are required since there is a desire to extend component lives to 50 000 h to minimise life-cycle costs. Similarly, for aeroengine applications the oxidation resistance of the bare alloy is assuming less importance due to the increased use of diffusion-type aluminide bond coats on which thermal barrier coatings are placed; nevertheless, oxidation resistance of the superalloy substrate will remain critical when the TBC system cannot be afforded as is still the case for many turbines.

6.2 Trends in superalloys and processes for turbine disc applications

Traditionally, superalloy turbine discs were designed to resist general yielding and low-cycle-fatigue cracking at sites of stress concentration near the bore; hence, great emphasis

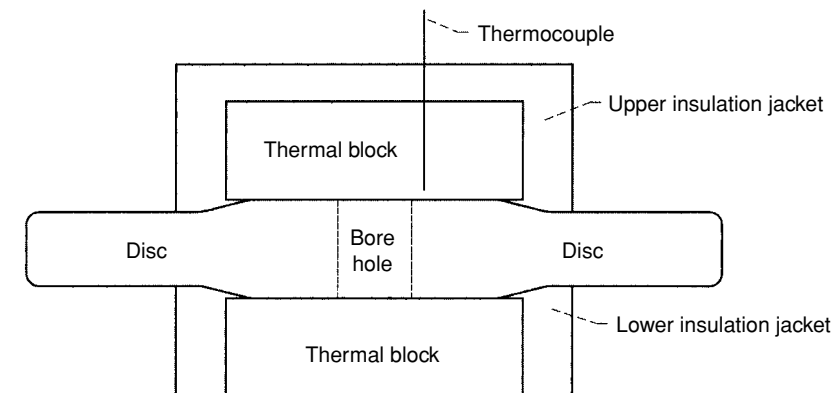


Fig. 6.6. Schematic representation of the disc, heat sinks and thermocouple arrangement used in the dual microstructure heat treatment process, as patented by NASA [4].

has been placed on imparting strengthening by grain size and precipitation hardening due to the γ' phase. But this situation has been altering steadily – particularly for high-pressure discs – as operating temperatures for the rim sections near the turbine blading, and thus the gas flow path, have been rising; in 2000 a figure of 750 °C or even 815 °C for high-performance military applications was possible for some advanced aeroengine designs. Consequently, the need to impart resistance to high-temperature ‘dwell’ crack propagation and creep is assuming greater importance. A corollary is that the requirements placed on the various processing steps required for component fabrication are becoming more stringent. Thirty years ago, the sole responsibility of the turbine disc supplier was to shape a superalloy billet – using a series of open/closed-die forming and machining operations – into a form suitable for heat treatment and ultrasonic inspection. Slowly, the requirements expanded to include demonstration of minimum property levels in both bore and rim regions. Nowadays, original equipment manufacturers (OEMs) are insisting on greater microstructural and property differentiation *within* any given component. Methods for this are now available; for example, the dual microstructure heat treatment technology (DMHT) technology designed and patented by NASA [4], in which both sub- and super-solvus heat treatments are carried out on bore and rim, respectively, *on a single component*. This is achieved by enhancing the thermal gradients present during heat treatment using heat sinks (typically solid metal cylinders) which chill the central portion of the disc as it resides in the furnace above the γ' -solvus; see Figure 6.6. This concept has been demonstrated for the state-of-the-art ME3 turbine disc superalloy (see Figure 6.7), with tensile and creep data which are superior in bore and rim regions, respectively, with experiments on a generic disc shape demonstrating the advantages and reliability of the dual grain structure at the component level [4]. Other propriety (and subtly different) methods are available to fabricate such ‘functionally graded’ discs, for example, General Electric’s dual heat treatment process, which was discussed in Chapter 4. At the time of writing, dual microstructure discs have yet to be employed in the gas turbine engine, but this situation is likely to change in the near future.

A consideration of powder metallurgy (P/M) superalloys – first introduced for high-pressure turbine discs in military engines, but now being employed increasingly in civil

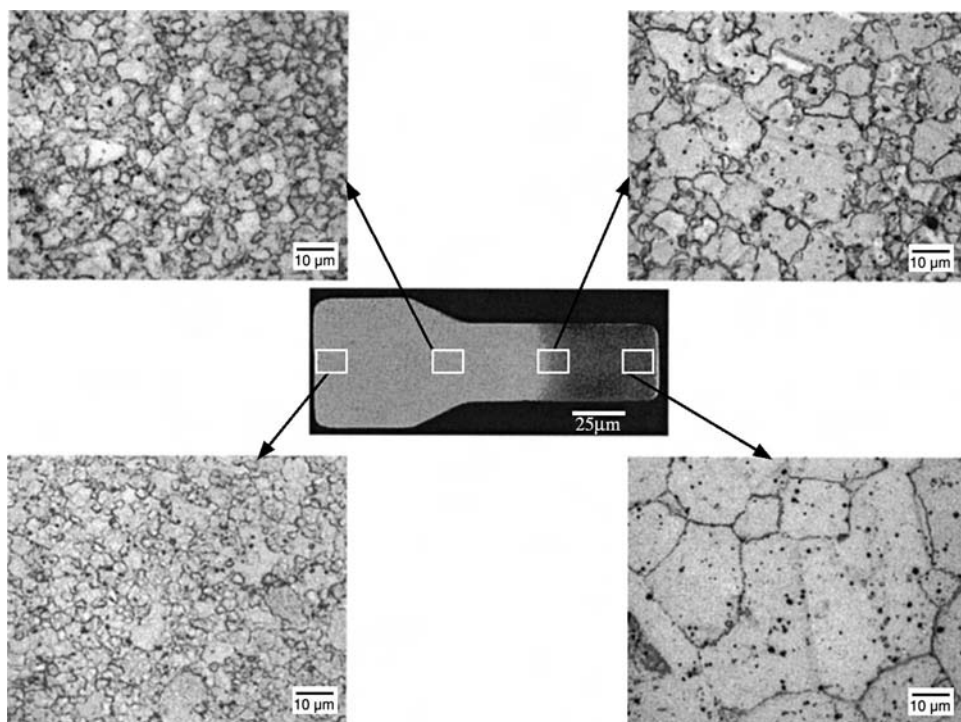


Fig. 6.7. Distribution of grain sizes in a ME3 turbine disc subjected to the dual microstructure heat treatment process [4].

aeroengines as stronger and more heavily alloyed variants are required – sheds further insight into the trends occurring. First-generation P/M superalloys (such as Rene 95) were designed with volume fractions of the gamma-prime (γ') phase which were high at around $\sim 55\%$ (Figure 4.17) and fine grain size of $\sim 10\ \mu\text{m}$, to achieve maximum tensile strength. Consequently, these were usually sub-solvus heat-treated. Second-generation P/M superalloys (such as Rene 88DT) were developed with lower γ' volume fractions and were heat-treated to larger grain sizes ($\sim 50\ \mu\text{m}$) to improve resistance to fatigue crack propagation; this was to ensure adequate life consistent with the damage-tolerance lifing methods. The third-generation alloys which are now being designed (such as ME3 and LSHR) have high γ' volume fractions as for the first-generation ones; where the DMHT process is not employed, grain sizes of $\sim 50\ \mu\text{m}$ are being sought so that dwell-crack growth and creep rates are minimised. The need for creep resistance in the rim regions has ramifications for the design of new powder disc compositions. Carbon and boron levels are being increased to improve creep behaviour and to promote precipitation of carbides and borides which provide pinning sites – to ensure that γ -grain growth does not occur beyond the solvus temperature in an abnormal fashion. As the use of P/M superalloys becomes more commonplace, the turbine industry will become increasingly dependent upon probabilistic methods of lifing. Consequently, better screening methods will be required for the rapid assessment of the quality of superalloy powder prior to its consolidation, so that distributions

of non-metallic inclusions can be quantified. To reduce the costs of powder material, increasing use will be made of hot isostatic pressing/conventional press conversion (cogging) practices rather than the hot compaction/extrusion routes – requiring specialised tooling and high tonnage equipment – which have been used traditionally for P/M billets [5]. For cast-and-wrought superalloy billets such as IN718 and Waspaloy, the possible presence of random melt-related defects will continue to prevent these being used to their full fatigue potential.

Process modelling of the operations used for disc production – particularly open- and closed-die forging, heat treatment and machining – is developing rapidly and will assume greater importance in the coming years. For the simulation of deformation trajectories and flow fields during forging operations, robust software packages based upon the finite-element method have been available for two decades or more; when underpinned by accurate data for the superalloy's thermal-viscoplastic flow, these have been used successfully for die design and for the prediction of defects such as cold shuts. More recently, thermal–elastic–plastic analyses of the heat-treatment process have been used increasingly for the prediction of residual stress fields driven by thermal gradients inherited from the quenching process [6]. Datasets of heat transfer coefficients and elastic/plastic constitutive behaviour are required. These allow the relationships between the heat transfer coefficient, part geometry, temperature field, material deformation behaviour and residual stress field to be quantified. Such modelling is enabling machining operations to be optimised to prevent excessive metal movement and deformation during lathing operations, which, if exceeding pre-defined tolerance levels, necessitate the scrapping of components. State-of-the-art process models make predictions of microstructure evolution, and this topic is currently an active area for research and development. There is evidence that the use of these process modelling tools is becoming increasingly necessary for quality assurance purposes. However, a further advantage is that the physical insights gained from it are stimulating the re-engineering of these processes such that component quality can be enhanced. Consider the advanced heat-treatment methods which are now in active development [7]. Traditionally, turbine discs have been quenched in oil so that the heat transfer coefficient around the periphery of the part is then uniform; consequently, thermal gradients are set up which cause the formation of a residual stress field. Whilst this cannot be entirely eliminated with the re-engineered process, its magnitude can be reduced by cooling using computer-controlled high-velocity gas jets directed at the disc, such that the heat transfer coefficient varies around its periphery in an optimal way; see Figure 6.8(a). Typically, this involves enhancing the heat transport coefficient at the bore region of the part on account of its greater thermal mass; the heat transfer coefficients required for this are deduced from the modelling. The advantage is that heat is then removed more uniformly than for traditional oil quenching, so that the thermal gradients responsible for the residual stress field are reduced; (see Figures 6.8(b) and (c)). A typical arrangement for such optimised heat treatment has as many as 12 different gas-flow zones which can be controlled as a function of quenching time and part temperature [7]. Such technology will be used increasingly, probably in conjunction with dual microstructure heat-treatment furnaces in the next-generation state-of-the-art heat-treatment facilities.

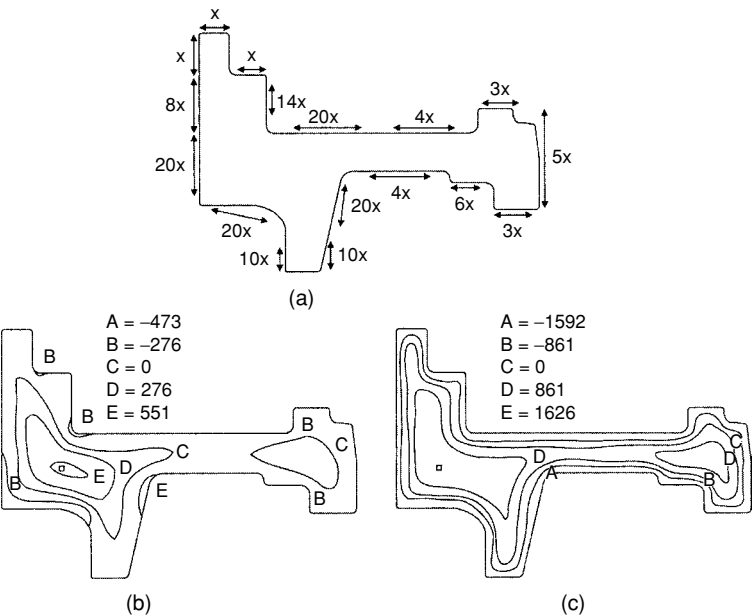


Fig. 6.8. Re-engineered process for the heat treatment of a turbine disc forging. (a) Variation of initial heat transfer coefficients optimised via process modelling. (b) Corresponding computed residual stress field and (c) computed residual stress field for the conventional oil-quench method [7]. Stress contours in (b) and (c) are given in units of megapascals.

6.3 Concluding remarks

Since the middle of the twentieth century, scientists and engineers have worked to develop the nickel-based superalloys and the processing methods required to put them to use in the gas turbine engine. Considerable progress has been made; this has been underpinned by a growing appreciation of the physical and chemical factors which govern their performance. As described in this book, the design of the alloys, the process development and the component engineering have not occurred independently – progress on these topics has tended to occur synergistically. At this stage of their development, the superalloys should rightly be regarded as one of the most complex of all the materials engineered by man. They represent a remarkable success story.

Nevertheless, there are reasons to believe that this class of material is not yet fully mature. Critical questions remain about the metallurgical factors which govern performance – for example, the creep strengthening provided by rhenium, the effect of ruthenium in imparting alloy stability and the role of reactive elements such as yttrium in improving oxidation performance. No well-developed theories exist for these effects. Furthermore, an adequate explanation has yet to be offered for the chemical effects which determine compatibility of the superalloys with thermal barrier coating systems. Bond coat chemistries have yet to be optimised. For the single-crystal superalloys, the effects of alloying additions on

the various creep degradation processes, for example, the rafting effect, have yet to be described in a quantitative sense. The factors controlling the creep performance of the disc alloys remain largely unexplored, and they provide an important topic for future research. With our understanding of these effects incomplete, it is unsurprising that most superalloy development work has tended to be highly empirical. Much emphasis has been placed on trial-and-error experimentation and screening trials, which have tended to be costly and time-consuming. Thus, whilst outstanding scientific questions remain, it is unlikely that the compositions of the superalloys have been fully optimised.

Significant challenges remain for present and future superalloy metallurgists. Most importantly, the physical metallurgy of the superalloys needs to be placed on a much firmer quantitative basis. Quantitative modelling will help in this regard, with important roles being played by modern electron theory, treatments of the kinetics of dislocation motion and process modelling tools which are sensitive to alloy chemistry. Since superalloy metallurgy is now in a highly developed state, the design of next-generation alloys will demand this. The widespread availability of computing hardware and modelling tools will help to achieve this goal.

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